# Anand Bhattad

Pixels · Perception · Physics

Assistant Professor of Computer Science · DSAI & Institute of Robotics
Johns Hopkins University

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## Research

Generative models look right without being right. My 3P Vision group builds the tools to tell the difference — to read out what they know and don't, and how to improve them — along three Ps:

**Pixels** — controllable generative models for relighting, compositing, and editing.

**Perception** — reading out the scene structure models implicitly know: intrinsics, geometry, and parts.

**Physics** — testing and enforcing physical and projective consistency, and carrying it to robots.

The group looks beyond generative models too: vision architectures, 3D perception, multisensory perception for embodied AI, and generative models for computational mechanics.

Keywords: Visual Generative Models; 3D Vision; Vision-based Robotics; Neural Rendering; Inverse Graphics; Intrinsic Images; Image-based Lighting; Physics Awareness in Large Vision Models.

## Education

**University of Illinois Urbana-Champaign (UIUC)**, USA

Ph.D., Computer Science 2019 – 2024

Thesis: **Exploring Knowledge in Generative Models**

Thesis committee: **David Forsyth (Advisor)**, Alexei A. Efros, William T. Freeman, Derek Hoiem, Svetlana Lazebnik, Shenlong Wang

M.S., Computer Science 2017 – 2018

M.S., Civil Engineering 2015 – 2017

**National Institute of Technology Karnataka (NITK) Surathkal**, India

B.Tech., Civil Engineering 2011 – 2015

## Work Experience

**Johns Hopkins University**, Assistant Professor of Computer Science; core member of DSAI & the Institute of Robotics 2025 – present

**Toyota Technological Institute at Chicago**, Research Assistant Professor 2023 – 2025

**University of California Berkeley**, Visiting Scholar; Host: Alexei A. Efros 2024

**Allen Institute for Artificial Intelligence (Ai2)**, Research Intern with Aniruddha Kembhavi 2023

**Intel**, Research Intern with Stephan R. Richter in Vladlen Koltun's team 2021

**NVIDIA**, Research Intern with Andrew Tao and Bryan Catanzaro 2020

**Fyusion Inc**, Research Intern with Abhishek Kar and Rodrigo Ortiz Cayon 2019

**Siemens Corporate Technology**, Computer Vision & DL Intern with Jan Ernst 2017

**Michigan State University**, Research Intern with Nizar Lajnef 2014

**IIIT Hyderabad**, Research Intern with Ramacharla Pradeep Kumar 2013

## Selected Honors and Awards

**Best Paper Award in Robot Learning**, ICRA 2026
**Outstanding Reviewer**, ICCV 2023
**Best Paper Finalist**, CVPR 2022
**Outstanding Emergency Reviewer**, CVPR 2021
**Excellent Teaching Assistant**, Intro Computing: Engrg & Sci 2016

## Selected Publications

*Self-selected publications (not necessarily the most cited work).*

- Generative Cinematographer: Composing Camera and Object Motion in 3D; **on arXiv**
- PowerSim: Differentiable Physics Simulation and Rendering with Power Diagrams; **on arXiv**
- Shadows Don't Lie and Lines Can't Bend! Generative Models Don't Know Projective Geometry...For Now; **CVPR 2024**
- StyleGAN Knows Normals, Depth, Albedo, and More; **NeurIPS 2023**
- PRINCIPIA: Relational Physics Tests for Video Models; **NeurIPS 2026 (Evaluations & Datasets Track)**
- Visual Jenga: Discovering Object Dependencies via Counterfactual Inpainting; **NeurIPS 2025**
- Latent Intrinsics Emerge from Training to Relight; **NeurIPS 2024** (**Spotlight**)
- Generative Blocks World: Moving Things Around in Pictures; **ICLR 2026**
- Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning; **ICRA 2026** (**Best Paper Award in Robot Learning**)
- LumiNet: Latent Intrinsics Meets Diffusion Models for Indoor Scene Relighting; **CVPR 2025**

## Advising

*Primary or co-advisor at Johns Hopkins University.*

**Postdoctoral:** Jamila Taaki *(primary advisor Laurent Pueyo)*.

**Doctoral:** Deming (Remus) Li *(co-advised with Rama Chellappa)*, Jiahan Zhang, Trong-Tung Nguyen.

**Master's & Undergraduate:** Chaohao Yang, Jiafeng Gu, Jin Hong (Jayden) Moon, Jiuzhou (Anthony) Cheng, Joanna Cheng, Rayhaneh Behravesh.

Complete publication list, invited talks, service, and advising at anandbhattad.github.io.